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Acoustic wave device

Abstract

An acoustic wave device includes a piezoelectric substrate, a first interdigital transducer electrode on the piezoelectric substrate, and a first reflector and a second reflector. The first interdigital transducer electrode, the first reflector, and the second reflector each include a plurality of electrode fingers. At least one of the first interdigital transducer electrode, the first reflector, and the second reflector has a nonuniform duty ratio area where three successive electrode fingers in an acoustic wave propagation direction all have different duty ratios.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2020-163568 filed on Sep. 29, 2020 and is a Continuation Application of PCT Application No. PCT/JP2021/034802 filed on Sep. 22, 2021. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an acoustic wave device.

2. Description of the Related Art

(2) Acoustic wave devices have been widely used, for example, in filters of mobile phones. Japanese Unexamined Patent Application Publication No. 2004-112591 discloses an example of a longitudinally coupled resonator-type acoustic wave filter serving as an acoustic wave device. This acoustic wave device includes a plurality of comb electrode arrays on a piezoelectric substrate. Each comb electrode array includes a pair of comb electrodes. One and the other comb electrodes of each comb electrode array have different duty ratios. This improves insertion loss.

SUMMARY OF THE INVENTION

(3) When an acoustic wave device, such as that described in Japanese Unexamined Patent Application Publication No. 2004-112591, is used, the occurrence of unwanted resonance may lead to degradation of filter characteristics. Particularly in a longitudinally coupled resonator-type acoustic wave filter configured to form an attenuation region using a longitudinal mode, the intensity of the longitudinal mode increases and a response level in the attenuation region may be degraded.

(4) Preferred embodiments of the present invention provide acoustic wave devices that each improve a response level in an attenuation region outside a pass band.

(5) An acoustic wave device according to a broad aspect of a preferred embodiment of the present invention includes a piezoelectric substrate, an interdigital transducer (IDT) electrode on the piezoelectric substrate, and a first reflector and a second reflector on the piezoelectric substrate and on both sides of the interdigital transducer electrode in an acoustic wave propagation direction. The interdigital transducer electrode, the first reflector, and the second reflector each include a plurality of electrode fingers. At least one of the interdigital transducer electrode, the first reflector, and the second reflector has a nonuniform duty ratio area where three successive electrode fingers in the acoustic wave propagation direction all have different duty ratios.

(6) An acoustic wave device according to another broad aspect of another preferred embodiment of the present invention includes a piezoelectric substrate, an interdigital transducer electrode on the piezoelectric substrate, and a first reflector and a second reflector on the piezoelectric substrate and on both sides of the interdigital transducer electrode in an acoustic wave propagation direction. The interdigital transducer electrode, the first reflector, and the second reflector each include a plurality of electrode fingers. When a value obtained by dividing a width of any electrode finger by an average width of all the electrode fingers of the interdigital transducer electrode, the first reflector, and the second reflector is defined as an average width ratio of the electrode finger, at least one of the interdigital transducer electrode, the first reflector, and the second reflector has a nonuniform average width ratio area where three successive electrode fingers in the acoustic wave propagation direction all have different average width ratios.

(7) The acoustic wave devices according to preferred embodiments of the present invention each improve the response level in the attenuation region outside the pass band.

(8) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a circuit diagram of a filter apparatus including an acoustic wave device according to a first preferred embodiment of the present invention.

(2) FIG. **2** is a schematic plan view of an area including a first reflector and a first interdigital transducer electrode of the acoustic wave device according to the first preferred embodiment of the present invention.
(3) FIG. **3** is a schematic elevational cross-sectional view of an area including three successive electrode fingers in the acoustic wave propagation direction, in the first interdigital transducer electrode according to the first preferred embodiment of the present invention.
(4) FIG. **4** is a diagram illustrating the duty ratio of each electrode finger of a plurality of interdigital transducer electrodes, the first reflector, and a second reflector in the first preferred embodiment of the present invention.
(5) FIG. **5** is a diagram illustrating attenuation-frequency characteristics of the filter apparatus including the acoustic wave device according to the first preferred embodiment of the present invention and a first comparative example.
(6) FIG. **6** is a diagram illustrating a relation between the standard deviation of the duty ratio and the response level outside the pass band of the filter apparatus including the acoustic wave device according to the first preferred embodiment of the present invention.
(7) FIG. **7** is a schematic plan view of an area including a first reflector and a first interdigital transducer electrode of an acoustic wave device according to a first modification of the first preferred embodiment of the present invention.
(8) FIG. **8** is a schematic plan view of an area including the first reflector and a first interdigital transducer electrode of an acoustic wave device according to a second modification of the first preferred embodiment of the present invention.
(9) FIG. **9** is a schematic elevational cross-sectional view of an area including three successive electrode fingers in the acoustic wave propagation direction, in the first interdigital transducer electrode according to a third modification of the first preferred embodiment of the present invention.
(10) FIG. **10** is a schematic plan view of an acoustic wave device according to a second preferred embodiment of the present invention.
(11) FIG. **11** is a diagram illustrating the average width ratio of each electrode finger of an interdigital transducer electrode, the first reflector, and the second reflector in the second preferred embodiment of the present invention.
(12) FIG. **12** is a diagram illustrating attenuation-frequency characteristics of a filter apparatus including the acoustic wave device according to the second preferred embodiment of the present invention and a second comparative example.
(13) FIG. **13** is a diagram illustrating a relation between the standard deviation of the average width ratio of an electrode finger and the response level of a longitudinal mode ripple in the filter apparatus including the acoustic wave device according to the second preferred embodiment of the present invention.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(14) The present invention will now be described by explaining preferred embodiments of the present invention with reference to the drawings.
(15) Note that the preferred embodiments described in the present specification are presented for illustrative purposes, and that some components described in different preferred embodiments can be replaced or combined.
(16) FIG. **1** is a circuit diagram of a filter apparatus including an acoustic wave device according to a first preferred embodiment of the present invention. FIG. **1** schematically illustrates an acoustic wave device **1**.
(17) The acoustic wave device **1** of the present preferred embodiment is a longitudinally coupled resonator-type acoustic wave filter. A filter apparatus **10** includes the acoustic wave device **1**. The filter apparatus **10** is a filter apparatus according to a preferred embodiment of the present invention. The filter apparatus **10** is a receiving filter. The filter apparatus **10** may be a transmitting

filter. A circuit configuration of the filter apparatus **10** is not particularly limited. The filter apparatus **10** is simply required to include the acoustic wave device **1** according to a preferred embodiment of the present invention.
(18) FIG. **2** is a schematic plan view of an area including a first reflector and a first interdigital transducer electrode of the acoustic wave device according to the first preferred embodiment of the present invention. No wires connected to the acoustic wave device **1** are shown in FIG. **2**. The same applies to the other plan views, as well as to FIG. **2**.
(19) The acoustic wave device **1** is a seven-interdigital-transducer longitudinally coupled resonator-type acoustic wave filter. The acoustic wave device **1** includes a piezoelectric substrate **2**. A plurality of interdigital transducer electrodes are disposed on the piezoelectric substrate **2**. Acoustic waves are excited by applying an alternating-current voltage to each of the interdigital transducer electrodes. In the present preferred embodiment, surface acoustic waves are excited in the acoustic wave device **1**. The plurality of interdigital transducer electrodes are arranged along the acoustic wave propagation direction. A pair of reflectors is disposed on the piezoelectric substrate and arranged on both sides of the plurality of interdigital transducer electrodes in the acoustic wave propagation direction. Specifically, the pair of reflectors includes a first reflector **4**A and a second reflector **4**B illustrated in FIG. **1**. The plurality of interdigital transducer electrodes include a first interdigital transducer electrode a second **3**A, interdigital transducer electrode **3**B, a third interdigital transducer electrode **3**C, a fourth interdigital transducer electrode **3**D, a fifth interdigital transducer electrode **3**E, a sixth interdigital transducer electrode **3**F, and a seventh interdigital transducer electrode **3**G arranged in this order from the first reflector **4**A. The number of interdigital transducer electrodes included in the acoustic wave device **1** is not limited to seven. For example, the acoustic wave device **1** may be of a three-interdigital-transducer type or five-interdigital-transducer type.
(20) As illustrated in FIG. **2**, the first interdigital transducer electrode **3**A includes a pair of busbars and a plurality of electrode fingers. The pair of busbars includes a first busbar **12** and a second busbar **13**. The first busbar **12** and the second busbar **13** are opposite each other. The plurality of electrode fingers of the first interdigital transducer electrode **3**A include a plurality of first electrode fingers **14** and a plurality of second electrode fingers **15**. The plurality of first electrode fingers **14** are each connected at one end thereof to the first busbar **12**. The plurality of second electrode fingers **15** are each connected at one end thereof to the second busbar **13**. The plurality of first electrode fingers **14** and the plurality of second electrode fingers **15** are interdigitated with each other. Like the first interdigital transducer electrode **3**A, the other interdigital transducer electrodes each include a pair of busbars and a plurality of electrode fingers.
(21) The first reflector **4**A includes a plurality of electrode fingers **16**. Similarly, the second reflector **4**B includes a plurality of electrode fingers. The interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the present preferred embodiment have the same electrode finger pitch. The electrode finger pitch refers to a center-to-center distance between adjacent electrode fingers. It is not necessarily required that all the electrode finger pitches of the interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B be the same.
(22) In the present specification, the duty ratio is defined for each electrode finger. That is, in each of the interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B, a value obtained by dividing the width of any electrode finger by the electrode finger pitch is defined as the duty ratio of the electrode finger. Specifically, $d=w/p$ is satisfied, where d is the duty ratio, w is the width of the electrode finger, and p is the electrode finger pitch. The width of the electrode finger is a dimension of the electrode finger along the acoustic wave propagation direction.
(23) An area where three successive electrode fingers in the acoustic wave propagation direction all have different duty ratios is defined as a nonuniform duty ratio area. In the present specification, the electrode finger pitch used to calculate the duty ratio is the average of center-to-center distances between any electrode finger and electrode fingers on both sides thereof. Details of the electrode

finger pitch and an example of the nonuniform duty ratio area will now be described with reference to FIG. **3**.
(24) FIG. **3** is a schematic elevational cross-sectional view of an area including three successive electrode fingers in the acoustic wave propagation direction, in the first interdigital transducer electrode according to the first preferred embodiment. An electrode finger **18**A is an electrode finger at one end of the first interdigital transducer electrode **3**A in the acoustic wave propagation direction. The electrode finger **18**A, an electrode finger **18**B, and an electrode finger **18**C are three successive electrode fingers in the acoustic wave propagation direction.
(25) The electrode finger pitch used to calculate the duty ratio of the electrode finger **18**B illustrated in FIG. **3** is the average of a center-to-center distance L**1** between the electrode finger **18**B and the electrode finger **18**A and a center-to-center distance L**2** between the electrode finger **18**B and the electrode finger **18**C. Specifically, the electrode finger pitch used to calculate the duty ratio of the electrode finger **18**B is equal to a distance A from the center between the center of the electrode finger **18**B and the center of the electrode finger **18**A to the center between the center of the electrode finger **18**B and the center of the electrode finger **18**C.
(26) On the other hand, the electrode finger pitch used to calculate the duty ratio of an electrode finger at one or the other end in the acoustic wave propagation direction is the center-to-center distance between the electrode finger and another electrode finger adjacent thereto. For example, the electrode finger pitch used to calculate the duty ratio of the electrode finger **18**A illustrated in FIG. **3** is the center-to-center distance L**1** between the electrode finger **18**A and the electrode finger **18**B. The same applies to the first reflector **4**A and the second reflector **4**B.
(27) Although the electrode finger pitches are uniform in the area illustrated in FIG. **3**, the electrode finger **18**A, the electrode finger **18**B, and the electrode finger **18**C all have different widths. This means that the illustrated area is a nonuniform duty ratio area.
(28) One of the unique features of the present preferred embodiment is that at least one of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B has a nonuniform duty ratio area. This can improve the response level in the attenuation region outside the pass band. Details of this effect will now be described along with details of the present preferred embodiment.
(29) FIG. **4** is a diagram illustrating the duty ratio of each electrode finger of the plurality of interdigital transducer electrodes, the first reflector, and the second reflector in the first preferred embodiment. The electrode finger number represented by the horizontal axis of FIG. **4** increases in the direction from the first reflector **4**A toward the second reflector **4**B.
(30) In the present preferred embodiment, the duty ratios in the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B are random. The entire area of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B is a nonuniform duty ratio area. In the acoustic wave device **1**, the minimum duty ratio is about 0.39, the maximum duty ratio is about 0.62, and the average duty ratio is about 0.5, for example. In the acoustic wave device **1**, about 95% of the duty ratios of all the electrode fingers are greater than or equal to about 0.45 and less than or equal to about 0.55, for example. In the acoustic wave device **1**, the standard deviation in the distribution of all the duty ratios in the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B is about 0.024, for example.
(31) The circuit configuration of the filter apparatus **10** according to the present preferred embodiment is as illustrated in FIG. **1**. Specifically, the filter apparatus **10** includes a first signal terminal **5** and a second signal terminal **6**. The first signal terminal **5** is an antenna terminal. The first signal terminal **5** is connected to an antenna. The first signal terminal **5** and the second signal terminal **6** may be configured as electrode pads, or may be configured as wires. The filter apparatus **10** further includes an acoustic wave resonator S**1**, an acoustic wave resonator S**2**, an acoustic wave resonator P**1**, and an acoustic wave resonator P**2**. The acoustic wave resonator S**1** and the acoustic

wave resonator S**2** are series traps. The acoustic wave resonator S**1** is connected between the first signal terminal **5** and the acoustic wave device **1**. The acoustic wave resonator S**2** is connected between the acoustic wave device **1** and the second signal terminal **6**. The acoustic wave resonator P**1** and the acoustic wave resonator P**2** are parallel traps. The acoustic wave resonator P**1** is connected between a ground potential and a node between the acoustic wave resonator S**1** and the acoustic wave device **1**. The acoustic wave resonator P**2** is connected between the second signal terminal **6** and the ground potential.
(32) In the present preferred embodiment, as described above, it is possible to improve the response level in the attenuation region outside the pass band. This effect will now be described by comparing the filter apparatus **10** with the first comparative example. A filter apparatus according to the first comparative example differs from the present preferred embodiment in that the duty ratios in each of a plurality of interdigital transducer electrodes, a first reflector, and a second reflector of a longitudinally coupled resonator-type acoustic wave filter are uniform.
(33) FIG. **5** is a diagram illustrating attenuation-frequency characteristics of the filter apparatus including the acoustic wave device according to the first preferred embodiment and the first comparative example.
(34) As indicated by arrow B**1**, arrow B**2**, and arrow B**3** in FIG. **5**, the response level corresponding to the unwanted wave outside the pass band is high in the first comparative example. In the first preferred embodiment, on the other hand, the response level corresponding to the unwanted wave is low. In particular, as indicated by arrow B**1**, the response level in the attenuation region outside the pass band can be improved in the first preferred embodiment.
(35) In the first comparative example, where the duty ratios are uniform, signals reflected from adjacent electrode fingers are in phase and this may cause unwanted resonance. In the first preferred embodiment having a nonuniform duty ratio area, however, signals reflected from adjacent electrode fingers in the nonuniform duty ratio area are less likely to be in phase. This lowers the Q factor related to the resonance of an unwanted wave, such as a longitudinal mode, indicated by arrow B**1**. Therefore, it is possible to suppress unwanted resonance and improve the response level corresponding to the unwanted wave.
(36) In the filter apparatus **10**, the standard deviation in the distribution of all the duty ratios in the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **1** was varied to measure the response level outside the pass band. The response level was measured at the frequency corresponding to arrow B**1** in FIG. **5**.
(37) FIG. **6** is a diagram illustrating a relation between the standard deviation of the duty ratio and the response level outside the pass band of the filter apparatus including the acoustic wave device according to the first preferred embodiment.
(38) FIG. **6** shows that when the standard deviation of the duty ratio is greater than or equal to about 0.015, the response level corresponding to a longitudinal mode, which is an unwanted wave, is lower than that when the standard deviation is less than about 0.015, for example. Therefore, it is preferable that the standard deviation in the distribution of all the duty ratios in the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **1** be greater than or equal to about 0.015, for example. This can further improve the response level in the attenuation region outside the pass band.
(39) It is preferable that the standard deviation in the distribution of all the duty ratios in the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **1** be less than or equal to about 0.55, for example. This makes it less likely that filter characteristics, such as insertion loss, will be degraded.
(40) In the distribution of all the duty ratios in the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **1**, greater than or equal to about 95% of all the duty ratios are preferably within the range of d.sub.ave±about 0.05, where d.sub.ave is the average duty ratio, for example. This makes it less likely that filter characteristics,

such as insertion loss, will be degraded.

(41) In the present preferred embodiment, as described above, the duty ratios in the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B are random. It is preferred, however, that at least one of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B have a nonuniform duty ratio area. This can still improve the response level in the attenuation region outside the pass band.

(42) In the present preferred embodiment, the interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B have the same electrode finger pitch. However, the electrode fingers of the interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B have random widths. This makes the duty ratios random. A value obtained by dividing the width of any electrode finger by the average width of all the electrode fingers of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B is defined as the average width ratio of the electrode finger. Specifically, wc=w/w.sub.ave is satisfied, where wc is the average width ratio, w is the width of the electrode finger, and w.sub.ave is the average width of electrode fingers. An area where three successive electrode fingers in the acoustic wave propagation direction all have different average width ratios is defined as a nonuniform average width ratio area. In the acoustic wave device **1**, the entire area of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B is a nonuniform average width ratio area. It is preferred, however, that at least one of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B have a nonuniform average width ratio area. This can still improve the response level in the attenuation region outside the pass band. It is not necessarily required that all the electrode finger pitches of the interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B be the same.

(43) As in the distribution of duty ratios described above, it is preferable that the standard deviation in the distribution of the average width ratios of all the electrode fingers of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **1** be greater than or equal to about 0.015, for example. This can further improve the response level in the attenuation region outside the pass band. It is preferable that the standard deviation in the distribution of the average width ratios of all the electrode fingers of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **1** be less than or equal to about 0.55, for example. This makes it less likely that filter characteristics, such as insertion loss, will be degraded.

(44) In the distribution of the average width ratios of all the electrode fingers of the plurality of interdigital transducer electrodes, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **1**, greater than or equal to about 95% of all the average width ratios are preferably within the range of about 1±0.05, for example. This makes it less likely that filter characteristics, such as insertion loss, will be degraded.

(45) FIG. **7** is a schematic plan view of an area including a first reflector and a first interdigital transducer electrode of an acoustic wave device according to a first modification of the first preferred embodiment.

(46) In the present modification, a first interdigital transducer electrode **23**X has an area where duty ratios are uniform and a nonuniform duty ratio area C. Specifically, in the first interdigital transducer electrode **23**X, the duty ratios are uniform in the entire area except the nonuniform duty ratio area C. The first interdigital transducer electrode **23**X may have a plurality of nonuniform duty ratio areas C. In this case, the areas where duty ratios are uniform and the nonuniform duty ratio areas C are alternately arranged. It is simply required that there be three or more electrode fingers in the nonuniform duty ratio area C. In the present modification, the duty ratios in a plurality of interdigital transducer electrodes other than the first interdigital transducer electrode **23**X, a first reflector **24**A, and a second reflector are uniform.

(47) In the present modification, the nonuniform duty ratio area C is also a nonuniform average

width ratio area. The first interdigital transducer electrode **23**X has an area where the widths of electrode fingers are uniform and the nonuniform average width ratio area. The first interdigital transducer electrode **23**X may have a plurality of nonuniform average width ratio areas. In this case, the areas where the widths of electrode fingers are uniform and the nonuniform average width ratio areas are alternately arranged. It is simply required that there be three or more electrode fingers in the nonuniform average width ratio area. In the present modification, the widths of electrode fingers in the plurality of interdigital transducer electrodes other than the first interdigital transducer electrode **23**X, the first reflector **24**A, and the second reflector are uniform. In the present modification, it is also possible to improve the response level in the attenuation region outside the pass band.
(48) FIG. **8** is a schematic plan view of an area including the first reflector and a first interdigital transducer electrode of an acoustic wave device according to a second modification of the first preferred embodiment.
(49) In the present modification, a first interdigital transducer electrode **23**Y includes a first region D, a second region E, and a third region F. The first region D is a region located at one end in the acoustic wave propagation direction. The second region E is a region located at the other end in the acoustic wave propagation direction. The third region F is a region adjacent to both the first region D and the second region E. The first region D and the second region E each include a plurality of electrode fingers. The electrode finger pitches in the first region D and the second region E are narrower than the electrode finger pitch in the other region.
(50) In the present modification, the nonuniform duty ratio area C and the nonuniform average width ratio area include the electrode finger in the center of the first interdigital transducer electrode **23**Y. Neither the nonuniform duty ratio area C nor the nonuniform average width ratio area is located at the boundary between the first region D and the third region F. Similarly, neither the nonuniform duty ratio area C nor the nonuniform average width ratio area is located at the boundary between and the second region E and the third region F. In the present modification, it is simply required that neither the nonuniform duty ratio area C nor the nonuniform average width ratio area be located at the boundaries described above. The nonuniform duty ratio area C and the nonuniform average width ratio area are not necessarily required to include the electrode finger in the center of the first interdigital transducer electrode **23**Y. In the present modification, it is also possible to improve the response level in the attenuation region outside the pass band.
(51) As illustrated in FIG. **3**, in the present preferred embodiment, the piezoelectric substrate **2** is a piezoelectric substrate including only a piezoelectric layer. Examples of the material that can be used to form the piezoelectric layer include lithium tantalate, lithium niobate, zinc oxide, aluminum nitride, crystal, and lead zirconate titanate (PZT). The piezoelectric substrate **2** may be a multilayer substrate including a piezoelectric layer or layers.
(52) FIG. **9** is a schematic elevational cross-sectional view of an area including three successive electrode fingers in the acoustic wave propagation direction, in the first interdigital transducer electrode according to a third modification of the first preferred embodiment.
(53) In the present modification, a piezoelectric substrate **22** includes a support substrate **25**, a high acoustic velocity film **26** serving as a high acoustic velocity material layer, a low acoustic velocity film **27**, and a piezoelectric layer **28**. Specifically, the high acoustic velocity film **26** is disposed on the support substrate **25**. The low acoustic velocity film **27** is disposed on the high acoustic velocity film **26**. The piezoelectric layer **28** is disposed on the low acoustic velocity film **27**.
(54) The low acoustic velocity film **27** is a film with a relatively low acoustic velocity. Specifically, the acoustic velocity of bulk waves propagating in the low acoustic velocity film **27** is lower than the acoustic velocity of bulk waves propagating in the piezoelectric layer **28**. Examples of the material that can be used to form the low acoustic velocity film **27** include glass, silicon oxide, silicon oxynitride, lithium oxide, tantalum pentoxide, and a material mainly including a compound produced by adding fluorine, carbon, or boron to silicon oxide.

(55) The high acoustic velocity material layer is a layer with a relatively high acoustic velocity. Specifically, the acoustic velocity of bulk waves propagating in the high acoustic velocity material layer is higher than the acoustic velocity of acoustic waves propagating in the piezoelectric layer **28**. Examples of the material that can be used to form the high acoustic velocity material layer include silicon, aluminum oxide, silicon carbide, silicon nitride, silicon oxynitride, sapphire, lithium tantalate, lithium niobate, crystal, alumina, zirconia, cordierite, mullite, steatite, forsterite, magnesia, a diamond-like carbon (DLC) film, and a medium such as diamond mainly including any of the materials described above.
(56) Examples of the material that can be used to form the support substrate **25** include piezoelectric materials, such as aluminum oxide, lithium tantalate, lithium niobate, and crystal; various ceramics, such as alumina, sapphire, magnesia, silicon nitride, aluminum nitride, silicon carbide, zirconia, cordierite, mullite, steatite, and forsterite; dielectrics, such as diamond and glass; semiconductors, such as silicon and gallium nitride; and resin.
(57) The piezoelectric substrate **22** of the present modification includes a stack of the high acoustic velocity material layer, the low acoustic velocity film **27**, and the piezoelectric layer **28**. This can effectively confine acoustic wave energy on the side of the piezoelectric layer **28**, and can also improve the response level in the attenuation region outside the pass band, as in the first preferred embodiment.
(58) The high acoustic velocity material layer may be a high acoustic velocity support substrate. In this case, the piezoelectric substrate may be a multilayer substrate including the high acoustic velocity support substrate, the low acoustic velocity film **27**, and the piezoelectric layer **28**. In the third modification, the piezoelectric layer **28** is indirectly disposed on the high acoustic velocity material layer, with the low acoustic velocity film **27** interposed therebetween. The piezoelectric layer **28** may be directly disposed on the high acoustic velocity material layer. The piezoelectric substrate may be a multilayer substrate without the low acoustic velocity film **27**. In this case, the piezoelectric substrate may be a multilayer substrate including the high acoustic velocity support substrate and the piezoelectric layer **28**. The piezoelectric substrate may be a multilayer substrate including the support substrate **25**, the high acoustic velocity film **26**, and the piezoelectric layer **28**. In any of the cases described above, it is still possible to effectively confine acoustic wave energy on the side of the piezoelectric layer **28**, and to improve the response level in the attenuation region outside the pass band.
(59) The piezoelectric layer **28** and an acoustic reflective film may define a layered body. The acoustic reflective film includes at least one low acoustic impedance layer and at least one high acoustic impedance layer. The low acoustic impedance layer is a layer with relatively low acoustic impedance. The high acoustic impedance layer is a layer with relatively high acoustic impedance. The low and high acoustic impedance layers are alternately stacked. This can also effectively confine acoustic wave energy on the side of the piezoelectric layer **28**, and can improve the response level in the attenuation region outside the pass band, as in the first preferred embodiment.
(60) In the first preferred embodiment and the modifications thereof described above, the acoustic wave device is a longitudinally coupled resonator-type acoustic wave filter, for example. Alternatively, an acoustic wave device according to a preferred embodiment of the present invention may be an acoustic wave resonator. This example will now be described in a second preferred embodiment.
(61) FIG. **10** is a schematic plan view of an acoustic wave device according to the second preferred embodiment.
(62) An acoustic wave device **31** is an acoustic wave resonator. The acoustic wave device **31** includes one interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B. The duty ratios in the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B are random. In the present preferred embodiment, the entire area of the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B is a nonuniform duty

ratio area. The electrode fingers of the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B have random widths. The entire area of the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B is a nonuniform average width ratio area. It is simply required, however, that at least one of the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B have a nonuniform duty ratio area or a nonuniform average width ratio area.
(63) In the present preferred embodiment, the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B have the same electrode finger pitch. It is not necessarily required that all the electrode finger pitches of the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B be the same.
(64) FIG. **11** is a diagram illustrating the average width ratio of each electrode finger of the interdigital transducer electrode, the first reflector, and the second reflector in the second preferred embodiment. The electrode finger number represented by the horizontal axis of FIG. **11** increases in the direction from the first reflector **4**A toward the second reflector **4**B. Numbers **1** to **11** represent the respective electrode f fingers of the first reflector **4**A, numbers **12** to **92** represent the respective electrode fingers of the interdigital transducer electrode **33**, and numbers **93** to **103** represent the respective electrode fingers of the second reflector **4**B.
(65) As illustrated in FIG. **11**, the average width ratios in the present preferred embodiment are greater than or equal to about 0.95 and less than or equal to about 1.05, for example. The standard deviation in the distribution of the average width ratios of all the electrode fingers of the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B of the acoustic wave device **31** is about 0.024, for example. The design parameters of the acoustic wave device **31** are as listed below. As viewed in the acoustic wave propagation direction, a region where adjacent electrode fingers of the interdigital transducer electrode overlap is defined as an overlap region. A dimension along the direction in which a plurality of electrode fingers in the overlap region extend is defined as an overlap width. A center-to-center distance between an electrode finger at an end portion of the interdigital transducer electrode in the acoustic wave propagation direction and an electrode finger of a reflector closest to the interdigital transducer electrode is defined as an I-R gap. Number of pairs of electrode fingers of interdigital transducer electrode **33**: 40.5; Overlap width of interdigital transducer electrode **33**: about 50 μm; Number of electrode fingers of first reflector **4**A and second reflector **4**B: 11; Electrode finger pitch of interdigital transducer electrode **33**, first reflector **4**A, and second reflector **4**B: about 4 μm; and I-R gap: about 4 μm.
(66) In the present preferred embodiment, it is possible, as in the first preferred embodiment, to improve the response level in the attenuation region outside the pass band. This will be described by comparing a filter apparatus including the acoustic wave device **31** of the present preferred embodiment with a second comparative example. A filter apparatus of the second comparative example differs from the present preferred embodiment in that the duty ratios and the widths of electrode fingers are uniform in each of the interdigital transducer electrode, the first reflector, and the second reflector of the acoustic wave resonator.
(67) FIG. **12** is a diagram illustrating attenuation-frequency characteristics of the filter apparatus including the acoustic wave device according to the second preferred embodiment and the second comparative example. Arrow G in FIG. **12** indicates a longitudinal mode ripple.
(68) FIG. **12** shows that the longitudinal mode ripple is more suppressed in the second preferred embodiment than in the second comparative example. This means that the response caused by the longitudinal mode ripple can be suppressed in the second preferred embodiment. Accordingly, for example, when a filter apparatus or a multiplexer includes the acoustic wave device **31** of the second preferred embodiment, the response level in the attenuation region outside the pass band can be improved.
(69) The standard deviation in the distribution of the average width ratios of all the electrode fingers of the interdigital transducer electrode **33**, the first reflector **4**A, and the second reflector **4**B

of the acoustic wave device **31** was varied to measure the response level outside the pass band. The response level was measured at the frequency corresponding to arrow G in FIG. **12**.
(70) FIG. **13** is a diagram illustrating a relation between the standard deviation of the average width ratio of the electrode finger and the response level of the longitudinal mode ripple in the filter apparatus including the acoustic wave device according to the second preferred embodiment. FIG. **13** shows that the longitudinal mode is more suppressed at a higher position in the vertical axis.
(71) FIG. **13** shows that when the standard deviation of the average width ratio of the electrode finger is greater than or equal to about 0.015, the longitudinal mode ripple is more suppressed than when the standard deviation is less than about 0.015, for example. Therefore, the response level in the attenuation region outside the pass band can be further improved in this case.
(72) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An acoustic wave device comprising: a piezoelectric substrate; an interdigital transducer electrode on the piezoelectric substrate; and a first reflector and a second reflector on the piezoelectric substrate and on both sides of the interdigital transducer electrode in an acoustic wave propagation direction; wherein the interdigital transducer electrode, the first reflector, and the second reflector each include a plurality of electrode fingers; at least one of the interdigital transducer electrode, the first reflector, and the second reflector has a nonuniform duty ratio area where three successive electrode fingers in the acoustic wave propagation direction all have different duty ratios; and a standard deviation in a distribution of all duty ratios in the interdigital transducer electrode, the first reflector, and the second reflector is greater than or equal to about 0.015.
2. A filter apparatus comprising: the acoustic wave device according to claim 1.
3. A filter apparatus according to claim 2, wherein the filter apparatus is a receiving filter.
4. A filter apparatus according to claim 2, wherein the filter apparatus is a transmitting filter.
5. The acoustic wave device according to claim 1, wherein the interdigital transducer electrode, the first reflector, and the second reflector each have an area where duty ratios are uniform or substantially uniform.
6. The acoustic wave device according to claim 1, wherein the acoustic wave device is an acoustic wave resonator including one interdigital transducer electrode.
7. The acoustic wave device according to claim 1, wherein the acoustic wave device is a longitudinally coupled resonator acoustic wave filter including a plurality of interdigital transducer electrodes; and the plurality of interdigital transducer electrodes are positioned along the acoustic wave propagation direction.
8. The acoustic wave device according to claim 1, wherein electrode finger pitches in the nonuniform duty ratio area are equal or substantially equal.
9. An acoustic wave device comprising: a piezoelectric substrate; an interdigital transducer electrode on the piezoelectric substrate; and a first reflector and a second reflector on the piezoelectric substrate and on both sides of the interdigital transducer electrode in an acoustic wave propagation direction; wherein the interdigital transducer electrode, the first reflector, and the second reflector each include a plurality of electrode fingers; when a value obtained by dividing a width of any of the electrode fingers by an average width of all the electrode fingers of the interdigital transducer electrode, the first reflector, and the second reflector is defined as an average width ratio of the electrode finger, at least one of the interdigital transducer electrode, the first reflector, and the second reflector has a nonuniform average width ratio area where three

successive electrode fingers in the acoustic wave propagation direction all have different average width ratios; and a standard deviation in a distribution of widths of all the electrode fingers of the interdigital transducer electrode, the first reflector, and the second reflector is greater than or equal to about 0.015.

10. A filter apparatus comprising: the acoustic wave device according to claim 9.

11. A filter apparatus according to claim 10, wherein the filter apparatus is a receiving filter.

12. A filter apparatus according to claim 10, wherein the filter apparatus is a transmitting filter.

13. The acoustic wave device according to claim 9, wherein the acoustic wave device is an acoustic wave resonator including one interdigital transducer electrode.

14. The acoustic wave device according to claim 9, wherein the acoustic wave device is a longitudinally coupled resonator acoustic wave filter including a plurality of interdigital transducer electrodes; and the plurality of interdigital transducer electrodes are positioned along the acoustic wave propagation direction.

15. The acoustic wave device according to claim 9, wherein electrode finger pitches in the nonuniform average width ratio area are equal or substantially equal.

16. The acoustic wave device according to claim 9, wherein the interdigital transducer electrode, the first reflector, and the second reflector each have an area where duty ratios are uniform or substantially uniform.

17. An acoustic wave device comprising: a piezoelectric substrate; an interdigital transducer electrode on the piezoelectric substrate; and a first reflector and a second reflector on the piezoelectric substrate and on both sides of the interdigital transducer electrode in an acoustic wave propagation direction; wherein the interdigital transducer electrode, the first reflector, and the second reflector each include a plurality of electrode fingers; when a value obtained by dividing a width of any of the electrode fingers by an average width of all the electrode fingers of the interdigital transducer electrode, the first reflector, and the second reflector is defined as an average width ratio of the electrode finger, at least one of the interdigital transducer electrode, the first reflector, and the second reflector has a nonuniform average width ratio area where three successive electrode fingers in the acoustic wave propagation direction all have different average width ratios; the interdigital transducer electrode includes a first region including the electrode finger at one end of the interdigital transducer electrode in the acoustic wave propagation direction and a second region including the electrode finger at the other end of the interdigital transducer electrode in the acoustic wave propagation direction; and the nonuniform average width ratio area is located outside a boundary between the first region and a region adjacent to the first region and outside a boundary between the second region and a region adjacent to the second region.

18. An acoustic wave device comprising: a piezoelectric substrate; an interdigital transducer electrode on the piezoelectric substrate; and a first reflector and a second reflector on the piezoelectric substrate and on both sides of the interdigital transducer electrode in an acoustic wave propagation direction; wherein the interdigital transducer electrode, the first reflector, and the second reflector each include a plurality of electrode fingers; at least one of the interdigital transducer electrode, the first reflector, and the second reflector has a nonuniform duty ratio area where three successive electrode fingers in the acoustic wave propagation direction all have different duty ratios; the interdigital transducer electrode includes a first region including the electrode finger at one end of the interdigital transducer electrode in the acoustic wave propagation direction, a second region including the electrode finger at the other end of the interdigital transducer electrode in the acoustic wave propagation direction, and a third region adjacent to both the first region and the second region; and the nonuniform duty ratio area is located outside a boundary between the first region and the third region and outside a boundary between the second region and the third region.